



**SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES**



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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**CSA0908
PROGRAMMING IN JAVA**

ASSIGNMENT – 1

1. Write a program to find the number of composite numbers in an array of elements

Sample Input:

Array of elements = {16, 18, 27, 16, 23, 21, 19}

Output:

Number of Composite Numbers = 5 Test

cases:

1. Array of elements = {26, 28, 37, 26, 33, 31, 29}
2. Array of elements = {1.6, 1.8, 2.7, 1.6, 2.3, 2.1, .19}
3. Array of elements = {0, 160, 180, 270, 160, 230, 210, 190, 0}
4. Array of elements = {200, 180, 180, 270, 270, 270, 190, 200}
5. Array of elements = {100, 100, 100, 100, 100, 100, 100, 100}

The screenshot shows the SIMATS Online Programming Lab interface. The user is logged in as S K DEEPA ARATHI (192571020RB). The lab is for JAVA PROGRAMMING. The question is 'Q5 First Non-Repeated Character' (5 marks). The user has answered 5/5.

Your Code

```
1 import java.util.*;
2 public class Main {
3     private static boolean isComposite(long x) {
4         if (x < 4) return false;
5         if (x % 2 == 0) return x != 2;
6         for (long i = 3; i * i <= x; i += 2) {
7             if (x % i == 0) return true;
8         }
9         return false;
10    }
11    public static void main(String[] args) {
12        Scanner sc = new Scanner(System.in);
13        int n;
14        if (!sc.hasNextInt()) return;
15        n = sc.nextInt();
16        int count = 0;
17        for (int i = 0; i < n; i++) {
18            long x = sc.nextLong();
19            if (isComposite(x)) count++;
20        }
21        System.out.print("Number of Composite Numbers = " + count);
22    }
23 }
```

Input

```
16
18
27
16
..
```

Output

```
Exception in thread "main"
java.util.NoSuchElementException
at
java.base/java.util.Scanner.throw
For(Scanner.java:937)
at
..
```

Visible Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

Buttons: Run, Save

2. Write a program to print the numbers from M to N by skipping K numbers in between? Sample Input: M = 50

N = 100

K = 7

Sample Output:

50, 58, 66, 74,

Test cases:

1. M = 15, N = 05, K = 02
2. M = 25, N = 50, K = 04
3. M = 15, N = 100, K = -02
4. M = 0, N = 0, K = 2
5. M = 200, N = 200, K = 50

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, and a user profile for S K DEEPA ARATHI. The main area displays a problem statement: "Write a program to print the numbers from M to N by skipping K numbers in between. Sample: M=50 N=100 K=7". Below this is a "Your Code" section with a Java program that reads M, N, and K from input and prints the sequence of numbers from M to N, skipping K numbers at a time. The program uses a Scanner for input and a loop with a skip counter for the output. To the right of the code is an "Input" section with a text area containing "50", "100", and "7", and an "Output" section showing the result "50 58 66 74 82 90 98". At the bottom right, there is a "Visible Test Cases" section with three cases: Test Case 1, Test Case 2, and Test Case 3, each with a radio button.

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JAVA PROGRAMMING

S K DEEPA ARATHI
192571020RB

Composite Numbers in Array
5 marks

Q2
Skip K Numbers
5 marks

Q3
Reverse Alphabetical Order
5 marks

Q4
Palindrome Check
5 marks

Q5
First Non-Repeated Character
5 marks

5/5 answered

Write a program to print the numbers from M to N by skipping K numbers in between.
Sample: M=50 N=100 K=7

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         if (!sc.hasNextInt()) return;
6         int M = sc.nextInt();
7         int N = sc.nextInt();
8         int K = sc.nextInt();
9         int skip = 0;
10        boolean firstPrinted = true;
11        for (int x = M; x <= N; x++) {
12            if (skip == 0) {
13                if (!firstPrinted) System.out.print(" ");
14                System.out.print(x);
15                firstPrinted = false;
16                skip = K;
17            } else {
18                skip--;
19            }
20        }
21        sc.close();
22    }
23 }
```

Input

50
100
7

Output

50 58 66 74 82 90 98

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

3. Write a program to arrange the letters of the word alphabetically in reverse order
Sample Input:

Enter the word: MOSQUE

Sample Output:

Alphabetical Order: U S Q O M E Test

Case:

6. HYPOTHECATION
7. MATRICULATION
8. MANIPULATION

SIMATS Online Programming Lab
S K DEEPA ARATHI 192571020RB

Write a program to arrange the letters of the word alphabetically in reverse order.
Sample: MOSQUE -> U S Q O M E

Composite Numbers in Array 5 marks

Q2 Skip K Numbers 5 marks

Q3 Reverse Alphabetical Order 5 marks

Q4 Palindrome Check 5 marks

Q5 First Non-Repeated Character 5 marks

5/5 answered

Your Code

```

1 import java.io.*;
2 import java.util.*;
3 public class Main {
4     public static void main(String[] args) throws Exception {
5         BufferedReader br = new BufferedReader(new InputStreamReader(S
6         String s = br.readLine();
7         if (s == null) return;
8         s = s.trim();
9         char[] a = s.toCharArray();
10        for (int i = 0; i < a.length; i++) {
11            for (int j = i + 1; j < a.length; j++) {
12                if (a[j] > a[i]) {
13                    char tmp = a[i];
14                    a[i] = a[j];
15                    a[j] = tmp;
16                }
17            }
18        }
19        for (int i = 0; i < a.length; i++) {
20            if (i > 0) System.out.print(" ");
21            System.out.print(a[i]);
22        }
23    }

```

Input

MOSQUE

Output

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

4. Write a program using a choice to check Case 1: Given string is a palindrome or not
Case 2: Given number is palindrome or not Sample Input:

Case = 1 String =
MADAM Sample
Output:
Palindrome Test
cases:

1. MONEY
2. 5678765
3. MALAY12321ALAM
4. MALAYALAM
5. 1234.4321

SIMATS Online Programming Lab
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Write a program using a choice to check string/number palindrome.

Composite Numbers in Array 5 marks

Q2 Skip K Numbers 5 marks

Q3 Reverse Alphabetical Order 5 marks

Q4 Palindrome Check 5 marks

Q5 First Non-Repeated Character 5 marks

5/5 answered

Your Code

```

1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         String str = sc.nextLine();
8         String rev = "";
9
10        for (int i = str.length() - 1; i >= 0; i--) {
11            rev = rev + str.charAt(i);
12        }
13
14        if (str.equals(rev)) {
15            System.out.println("Palindrome");
16        } else {
17            System.out.println("Not Palindrome");
18        }
19    }
20 }

```

Input

SIMATS

Output

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

5. Java Program to Find the First Non-Repeated Character in a String.

Case 1: Find the first non-repeated character in a given string.

Case = 1

String = SWISS

First Non-Repeated Character = W

Test Cases

1. HELLO
2. SWISS
3. PROGRAMMING
4. AABBCDD
5. SUCCESS

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, and a user profile for S K DEEPA ARATHI. The main task is to find the first non-repeated character in a string, with a sample input of 'SWISS' and the expected output 'W'. The 'Your Code' section contains a Java program that uses a Scanner to read a string and a frequency array to find the first non-repeated character. The 'Input' field contains 'SWISS', and the 'Output' field is empty. The 'Visible Test Cases' section shows three test cases, all of which are currently failed.

SIMATS ENGINEERING SIMATS Online Programming Lab
JAVA PROGRAMMING

S K DEEPA ARATHI
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Find the first non-repeated character in a string.
Sample:SWISS->W

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         int[] freq = new int[256];
7         for (int i = 0; i < str.length(); i++) {
8             freq[(int) str.charAt(i)]++;
9         }
10        for (int i = 0; i < str.length(); i++) {
11            char ch = str.charAt(i);
12            if (freq[(int) ch] == 1) {
13                System.out.print(ch);
14                break;
15            }
16        }
17        sc.close();
18    }
19 }
```

Input
SWISS

Output

Visible Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

Run Save

Rubric

Criteria	Weight
Problem Understanding & Logic	20%
Program Execution & Output	30%
Algorithm & Efficiency	20%
Input Handling & Edge Cases	15%
Coding Standards & Documentation	15%

Rubrics Evaluation

Criteria	Wt. %	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Understanding	20	Correctly understands all requirements and handles all cases accurately.	Understands most requirements with minor omissions.	Partial understanding; several requirements are missed.	Incorrect understanding of the problem.
Algorithm Design	20	Develops an efficient and logically correct algorithm with proper flow.	Algorithm is mostly correct with minor logical issues.	Algorithm works only for basic cases and has logical errors.	Incorrect or incomplete algorithm.
Implementation using Java	25	Java program is syntactically correct, well-structured, and executes successfully for all test cases.	Minor syntax or logical issues; works for most test cases.	Multiple errors; works only for limited inputs.	Program does not compile or execute correctly.
Output & Test Case Validation	20	Produces correct output for sample input and all provided test cases, including edge cases.	Produces correct output for most test cases.	Correct for only a few test cases.	Incorrect output for most or all test cases.

Code Quality & Documentation	15	Well-formatted code with meaningful variable names, indentation, and comments.	Good formatting with few documentation issues.	Inconsistent formatting and limited comments.	Poor formatting, unclear code, and no comments.
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